Worked for 9m 31s

A. Verification verdict

REVISE — not yet PASS

Source control is verified:

> `% TRUTH VERSION: tbv_mdr_Paper01-v02.tex`

I reviewed the complete integrated 1,786-line file. It compiles successfully, and the source contains no missing internal labels.

B. Brief justification

The revision substantially improves the formal system:

- Irreparable stochastic records are now explicitly accommodated.
- Elections are correctly excluded when no restorative refinement exists.
- The positive production-success bound now applies only to repairable inquiries.
- Reference and acting beliefs resolve the active-set refill problem.
- The retained record is included in the SCUE state.
- The pinned structure and leverage statistic now have explicit mathematical definitions.

Nevertheless, the participation decision is not yet feasible under the agent's information, and parts of the persistence characterization remain incorrect as stated.

C. Issues preventing PASS

1. **The agent cannot compute the objects governing inquiry.**

   The agent observes only $(C_{\mathcal{P}_i}(s), a, x_i)$; the underlying s is retained only in analyst-level history (lines 456–465). Yet determining $\mathcal{R}$, the induced token partitions, Λ_i, and $\Theta_i(t)$ requires recoding tokens under every finer, presently unrepresented partition (lines 769–782 and 947–966).

   Thus two agent-indistinguishable records can differ in whether $\mathcal{R}$ is empty and in the resulting value of B_i. This conflicts with the claims that the agent recognizes irreparability and computes B_i before insight. The model needs either an explicit pre-insight oracle/cognitive primitive, richer observable records, or a construction based entirely on currently observable token conflicts.

2. **The stochastic persistence result is false under its stated premises.**

   Proposition 9(ii-b) treats a sufficiently large existing count as enough to make all future dissonance arbitrarily unlikely. Its Hoeffding argument, however, ignores the arbitrary pre-configuration prefix embedded in the cumulative empirical conditional.

   A record can have arbitrarily many observations while sitting exactly at the test boundary. The next observation can then empty the active set with a probability bounded away from zero, regardless of how large the count is. The result needs an entry-margin condition, a clean post-entry validation sample, or a weaker conclusion of eventual rather than uninterrupted stability.

3. **Exact SCUE persistence conflicts with empirical conjecture updating.**

   SCUE requires q_i to agree exactly with induced play, but the persistence results say the SCUE conditions hold in every subsequent generation while conjectures merely converge empirically. With nondegenerate induced play, finite empirical updating generally moves q_i away from exact equality. Either conjectures must be frozen, condition (ii) must be asymptotic, or the conclusion must be convergence to SCUE rather than continuous SCUE membership.

4. **The positive-leverage persistence example is inconsistent with the revised B_i.**

   Because Λ_i changes as the record grows, B_i need not “grow as evidence accumulates.” In the stated example, later pooled tokens can make the meet of the minimal repairs trivial, producing $B_2 = 0$, not the asserted $B_2 > 0$. Conversely, if a stable pinned rule continues earning a positive amount per recurrent token, the full-record sum eventually exceeds every fixed γ_2, forcing inquiry.

   The performance difference may therefore remain supportable through **zero leverage**, but the present claim of permanently positive leverage held below a fixed cost is not established.

A final local correction will also be required: the abstract, introduction, and conclusion say every agent eventually lives in a Bayesian world, whereas the theorem explicitly permits permanent dissonant stasis.